01.

Number System

Type - 1

1. Find the least number that is divisible by 12, 18, 21, and 30.

(a) 1060 (b) 1260
(c) 1620 (d) 1020

RRB JE - 25/05/2019 (Shift-III)

Ans : (b) LCM of the numbers 12, 18, 21 and 30 = 420
So the smallest number = 420 k
where k = 1, 2, 3, ----------
When taking the number k = 3
Number = 420 × 3 = 1260

2. A student divided a number by 12 instead of 21 and received 35. Find the correct answer.

(a) 20 (b) 15
(c) 26 (d) 25

RRB JE - 26/05/2019 (Shift-II)

Ans : (a) Let the number be x.
According to the question,
dividing by 12

$$\frac{x}{12} = 35$$

x = 35 × 12
x = 420
The number is 420
Dividing 420 by 21-

$$\frac{420}{21} = 20$$

Hence, the correct answer = 20

3. Find the least number to be added to 231228 to make it exactly divisible by 33.

(a) 3 (b) 4
(c) 2 (d) 1

RRB JE - 27/05/2019 (Shift-III)

Ans : (a)

```
        7006
33 ) 231228
     231
     ×××228
       198
        30
```

Hence, the required number = 33 – 30 = 3

4. Find the greatest number that will exactly divide 25, 35, 40 and 30.

(a) 35 (b) 15
(c) 20 (d) 5

RRB JE - 01/06/2019 (Shift-II)

Ans : (d) HCF of the numbers 25, 35, 30 and 40
25 = 5 × 5
35 = 5 × 7
40 = 5 × 8
30 = 5 × 6
HCF = 5
Hence, the required number will be 5.

5. Which of the following numbers is not divisible by 8?

(a) 12676 (b) 11504
(c) 12832 (d) 12360

RRB RPF Constable -24/01/2019 (Shift-II)

Ans : (a)

(a) $\frac{12676}{8} = 1584.5$

(b) $\frac{11504}{8} = 1438$

(c) $\frac{12832}{8} = 1604$

(d) $\frac{12360}{8} = 1545$

Hence, the number 12676 is not divisible by 8.
Note: Divisibility rule by 8- If the last three digits of a number is divisible by 8, the number will be divisible by 8.

6. 276x1, is divisible by 3. What is the sum of the possible values of x?

(a) 18 (b) 21
(c) 12 (d) 15

RRB RPF SI-12/01/2019 (Shift-I)

Ans. (d) If the sum of all digits of a number is divisible by 3, the number will be divisible by 3.
276x1, is divisible by 3.
2 + 7 + 6 + x + 1 = 16 + x
The number will be completely divisible by 3, by putting the possible values of x 2, 8, and 5.
Hence, the sum of the possible values of x = 2+8+5 =15

7. By dividing 14528 by a certain number, Suresh gets 83 as quotient and 3 as remainder. What is the divisor?

(a) 165 (b) 185
(c) 195 (d) 175

RRB RPF SI -06/01/2019 (Shift-III)

Ans : (d) Let the divisor is 'x'.
Given- Dividend = 14528
Quotient = 83
Remainder = 3
Dividend = (Divisor × Quotient + Remainder)

$\Rightarrow 14528 = (x \times 83) + 3$

$\Rightarrow 83x = 14528 - 3$

$\Rightarrow 83x = 14525$

$\Rightarrow x = \frac{14525}{83} \quad \Rightarrow x = 175$

8. Find the smallest number which, when added to 20000, the sum is divisible by 12,15 and 25.

(a) 180 (b) 150
(c) 200 (d) 100

RRB RPF Constable -17/01/2019 (Shift-I)

Ans : (d) LCM of 12, 15 and 25 = 300

$$\text{Remainder} = \frac{20000}{300} = 200$$

The required number which should be added to 2000

$= (300 - 200) = 100$

9. Find the smallest cube number more than 1000, that is divisible by 2, 4, and 6.

(a) 1296 (b) 5832
(c) 1728 (d) 4096

RRB RPF Constable -19/01/2019 (Shift-I)

Ans. (c) : LCM of 2, 4 and 6 = 12
Hence, the cube of the number 12 = $(12)^3$ = 1728

10. If the number X4461 is divisible by 11, find the value of x.

(a) 2 (b) 4
(c) 3 (d) 5

RRB Group-D – 17/09/2018 (Shift-I)

Ans : (d) Rule of divisibility by 11- if the difference between sum of even places of the number and the sum of digit odd places and the result is 0 or any number divisible by 11, then the number will also be divisible by 11.
The number – x 4 4 6 1

$x + 4 + 1 - (4 + 6) = 0$

$x + 5 - 10 = 0$

$x = 5$

11. Which number is divisible by 9?

(a) 56112 (b) 89445
(c) 49653 (d) 58556

RRB Group-D – 03/10/2018 (Shift-II)

Ans : (c) If the sum of all the digits of a number is divisible by 9, the number will also be completely divisible by 9.
Hence, from option-
(a) The sum of the digits of 56112 = 15 (×)
(b) The sum of the digits of 89445 = 30 (×)
(c) The sum of the digits of 49653 = 27 (√)
(d) The sum of the digits of 58556 = 29 (×)
Hence, the number divisible by 9 = 49653

12. Which of the following numbers is perfectly divisible by 4?

(a) 6542176 (b) 7253566
(c) 5632654 (d) 4187290

RRB Group-D – 17/09/2018 (Shift-III)

Ans. (a) : Rule of divisibility by 4-
If the number formed with last two digit of a number, is divisible by 4, the whole number will be divisible by 4.
From option (a)-

$$\begin{array}{r} 19 \\ 4\overline{)76} \\ 4 \\ \hline 36 \\ 36 \\ \hline \times\times \end{array}$$

13. If a natural number is divided by 4, 5, 6 or 7, leaves remainder 1 in each case. Find that smallest number.

(a) 421 (b) 61
(c) 841 (d) 211

RRB Paramedical Exam – 20/07/2018 (Shift-II)

Ans : (a) LCM of 4,5,6 and 7 = 420
The required number = 420 + 1 = 421
Hence, each condition 421 leaves remainder 1, when divided by 4, 5, 6 and 7.

14. Which of the following is the largest such number, which leaves remainder 9 and 20, when divides 105 and 164 respectively?

(a) 36 (b) 48
(c) 24 (d) 96

RRB Group-D – 22/10/2018 (Shift-II)

Ans : (b) 105 – 9 = 96 and 164 – 20 = 144

$$\begin{array}{r} 1 \\ 96\overline{)144} \\ 96 \\ 48\overline{)96}(2 \\ 96 \\ \hline \times\times \end{array}$$

Hence, so that largest number = HCF of 96 and 144 = 48

15. Which is the largest such number which leaves remainder 2 and 3, when divides 258 and 323 respectively?

(a) 40 (b) 24
(c) 64 (d) 132

RRB Group-D – 30/10/2018 (Shift-III)

Ans. (c) : 258 – 2 = 256
323 – 3 = 320
320 – 256 = 64 that is HCF
Hence, the required number is 64.
So dividing of 258 and 323 will give the remaining 2 and 3 respectively.

16. Which of the following numbers is divisible by 6?

(a) 12378 (b) 12363
(c) 12370 (d) 12388

RRB Group-D – 05/12/2018 (Shift-I)

Ans : (a) Divisibility rule by 6 – If a given number is divisible by 2 and 3, then the number will also be divisible by 6.
Divisibility rule by 2 – If the unit digit of a given number is divisible by 2, then the number will also be divisible by 2.
Divisibility rule by 3 – If the sum of all the digits of the number is divisible by 3, then the number will also be divisible by 3.
From option (a)- $1 + 2 + 3 + 7 + 8 = 21$

$$= \frac{21}{3} = 7$$

Hence, the number 12378 is divisible by 6.

17. Choose the missing digit 'x' from the options given for the number 987x54, so that the number is completely divisible by 6.

(a) 2 (b) 5
(c) 3 (d) 1

RRB Group-D – 18/09/2018 (Shift-I)

Ans. (c) : The given number will be divisible by 6 if it is divisible by 2 and 3.
Divisibility rule by 2 – If the unit digit is divisible by 2, then the number will also be divisible by 2.
Divisibility rule by 3 – If the sum of all the digits of the given number is divisible by 3, then the number will also be divisible by 3.
option (c) is

$$\Rightarrow \frac{9+8+7+x+5+4}{3} = \frac{33+x}{3}$$

Hence, the value of x will be 3.

18. What number should be deducted from 1265 to make it divisible by 29 exactly?

(a) 15 (b) 16
(c) 18 (d) 17

RRB NTPC 05.04.2016 Shift : 3

Ans : (c) To get the number divide 1265 by 29.

```
      43
  29)1265
     116
     ×105
      87
      18
```

Hence, 18 should be subtracted out of 1265 to make it completely divisible by 29.

19. What number, from the following, should be deducted from 1184 to make it exactly divisible be 21?

(a) 15 (b) 12
(c) 8 (d) 7

RRB NTPC 31.03.2016 Shift : 2

Ans.: (c) To get the required number divide 1184 by 21.

```
      56
  21)1184
     105
     ×134
      126
      ××8
```

Hence, 8 should be subtracted from 1184 to make it exactly divisible by 21.

20. Find the least number to be added to 1739 so that it is exactly divisible by 11.

(a) 11 (b) 2
(c) 1 (d) 10

RRB NTPC 30.03.2016 Shift : 1

Ans : (d) To get the required number divide 1739 by 11 then deduce the remainder from the divisor.

```
      158
  11)1739
     11
     ×63
      55
      ×89
       88
       ×1
```

Hence, the required number will be 11 – 1 = 10.

21. Find the least number which should be added to 1456 so that it is divisible by 6, 5 and 4 without leaving a remainder.

(a) 6 (b) 61
(c) 44 (d) 16

RRB NTPC 18.01.2017 Shift : 1

Ans : (c) LCM of 6, 5 and 4 = 60
For the least number,

$$\frac{1456}{60} = 24\frac{16}{60}$$

Hence, the required number = 60 – 16 = 44
By adding 44 in 1456 we get 1500 that is divisible by 6, 5 and 4.

22. Find the remainder, when 3^{10} is divided by 7.

(a) 4 (b) 3
(c) 5 (d) 6

RRB NTPC 18.04.2016 Shift : 3

Ans : (a) $3^{10} = 3\times3\times3\times3\times3\times3\times3\times3\times3\times3$
$= 59049$

$\therefore \quad \frac{59049}{7} = 4$ remainder

23. Which of the following numbers is divisible by 12?

(a) 93412 (b) 63412
(c) 73412 (d) 83412

RRB ALP & Tec. (31-08-18 Shift-II)

Ans. (d) : The number which is divisible by 12, should be divisible by 3 and 4 also.
If the sum of all the digits of a number is divisible by 3, the number will also be divisible by 3.
If the last two digit of a number are divisible by 4, the number will also be divisible by 4.
From option (d),
Then 8+3+4+1+2 = 18, Which is divisible by 3.
The last 2-digit of the number are 12, Which is also divisible by 4.
Hence, the number 83412 is divisible by 12.

24. Which of the following numbers is divisible by 9?
(a) 56765 (b) 47862
(c) 54321 (d) 87654
RRB ALP & Tec. (30-08-18 Shift-III)

Ans : (b) If the sum of all the digits of a number is divisible by 9, the number will be divisible by 9.
Hence, from option (b),
47852 ⇒ 4+7+8+6+2 = 27, which is divisible by 9.
Hence option (b) will be true.

25. If $3x^2 + ax + 4$ is perfectly divisible by $x - 5$, then the value of a is:
(a) – 12 (b) – 5
(c) – 15.8 (d) – 15.6
RRB ALP & Tec. (09-08-18 Shift-II)

Ans : (c) According to the question,
$3x^2 + ax + 4 = 0$ --------- (i)
∵ equation (i), is divisible by ($x-5$)
Hence,
⇒ $x - 5 = 0$
⇒ $x = 5$
Putting the value of x in equation(i),
$3(5)^2 + a \times 5 + 4 = 0$
$75 + 5a + 4 = 0$
$5a = -79$
$a = -15.8$

Type - 2

26. The sum of three consecutive odd numbers is more than first number of it by 20. Find the largest number among them.
(a) 13 (b) 9
(c) 11 (d) 7
RRB JE - 28/06/2019 (Shift-III)

Ans. (c) Let the three consecutive odd numbers are x, x +2, x +4.
According to the question,
$x + x+2 + x+4 = x + 20$
$3x + 6 = x + 20$
$2x = 14$
$x = 7$
Hence, the required number = x + 4 = 7 + 4 = 11

27. If 3/11 < x/3 < 7/11, which of the following can be value of 'x'?
(a) 0.5 (b) 1
(c) 2 (d) 3
RRB JE - 23/05/2019 (Shift-I)

Ans : (b) From option,
When X = .5 then .272 < .166 < .636 (False)
When X = 1 then .272 < .333 < .636 (True)
When X = 2 then .272 < .666 < .636 (False)
When X = 3 then .272 < 1 < .636 (False)
Hence, It is clear that the value of X will be 1.

28. The product of 4 consecutive numbers is always divisible by which of the following numbers?
(a) 10 (b) 22
(c) 24 (d) 48
RRB RPF SI -05/01/2019 (Shift-I)

Ans : (c) Let 4 consecutive numbers are n, (n+1), (n+2) and (n+3) respectively.
According to the question,
The Product of four consecutive numbers
= n(n+1)(n+2)(n+3)
Where n = 1,2,3,…….
Putting n = 1,
Product,
= 1 (1+ 1) (1+ 2) (1 + 3)
= 1 × 2 × 3 × 4 = 24
Putting n = 2,
Product,
= 2×3×4×5 = 24 × 5 = 120
Hence, the product of 4 consecutive numbers is always divisible by 24.

29. If the first number and the second number is 25% and 50% more than the third number respectively, find the ratio between the first and second number.
(a) 5 : 6 (b) 2 : 1
(c) 6 : 5 (d) 1 : 2
RRB JE - 27/05/2019 (Shift-I)

Ans : (a) Let the third number be 100.
Then the first number = 125
And the second number = 150
First number : Second number = 125 : 150
= $\boxed{5:6}$

30. Three times the first of three consecutive odd integers is 3 more than two times the third. Find the third integer.
(a) 15 (b) 13
(c) 11 (d) 9
RRB JE - 26/06/2019 (Shift-III)

Ans : (a) Let three consecutive odd integers = x, x+2, x+4
$3x = 2(x + 4) +3$
$3x = 2x + 8 + 3$
$x = 11$
Hence, the third integer = x + 4 = 11 + 4 = 15

31. How many total tens digit in the calculation from series 1 to 99?

(a) 98 (b) 90
(c) 99 (d) 100

RRB RPF Constable -17/01/2019 (Shift-II)

Ans : (b) The number of tens from 1 to 10 = 1
The number of tens from 11 to 90 = 80
The number of tens from 91 to 99 = 9
Hence, the total number of tens from series 1 to 99
= 1 + 80 + 9 = 90

32. Find the 10th number in the series 12, 19, 26, 33 ……..

(a) 89 (b) 75
(c) 82 (d) 68

RRB Group-D – 25/10/2018 (Shift-II)

Ans : (b) The given series 12, 19, 26, 33 …….. belongs to A.P.(Arithmetical progression)
Hence, the first term 'a' = 12, common difference 'd' = 19 – 12 = 7 and the number of terms 'n' = 10
Hence, the Nth term of the series,

$$T_n = a + (n-1)d$$

$$T_{10} = 12 + (10-1) \times 7 = 12 + 63 = 75$$

33. If the rational numbers $\frac{4}{-9}, \frac{-7}{18}, \frac{5}{-6}, \frac{-2}{3}$ are arranged in ascending order. Which number will be placed the first?

(a) $\frac{4}{-9}$ (b) $\frac{-7}{18}$
(c) $\frac{5}{-6}$ (d) $\frac{-2}{3}$

RRB Group-D – 02/11/2018 (Shift-II)

Ans. (c)
Given rational numbers = $\frac{4}{-9}, \frac{-7}{18}, \frac{5}{-6}$ and $\frac{-2}{3}$

$$\frac{4}{-9} = -0.44$$

$$\frac{-7}{18} = -0.38$$

$$\frac{5}{-6} = -0.83$$

$$\frac{-2}{3} = -0.66$$

The first number, when arranged in ascending order

$$= -0.83 = \frac{5}{-6}$$

34. How many terms in the series 7, 14, 21, 28, …… sum of 952?

(a) 16 (b) 17
(c) 18 (d) 19

RRB NTPC 28.04.2016 Shift : 1

Ans : (a) The given series- 7, 14, 21, 28, ……
Let the sum of n terms = 952
7 + 14 + 21 + 28 +……..= 952
⇒ 7 (1 + 2 + 3 + 4 +……) = 952
⇒ 1 + 2 + 3 + 4 +……= 136

$$\Rightarrow \frac{n(n+1)}{2} = 136$$

∵The sum of n consecutive numbers = $\frac{n(n+1)}{2}$

$\Rightarrow n^2 + n = 272$
$\Rightarrow n^2 + n - 272 = 0$
$\Rightarrow n^2 + 17n - 16n - 272 = 0$
$\Rightarrow (n - 16)(n + 17) = 0$
$n - 16 = 0 \Rightarrow n = 16$
Hence, the sum of 16 terms is 952.

35. Find two consecutive numbers where thrice the first number is more than twice the second number by 5.

(a) 5 and 6 (b) 6 and 7
(c) 7 and 8 (d) 9 and 10

RRB NTPC 28.03.2016 Shift : 1

Ans : (c) Let the consecutive numbers be x and x+1.
According to the question-

$$3x = 2(x+1) + 5$$

$$\Rightarrow 3x = 2x + 7$$

$$\Rightarrow x = 7$$

Hence, the required consecutive numbers will be 7 and 8.

36. Find the value of 6 + 11 + 16 + 21 + …. + 71

(a) 539 (b) 561
(c) 661 (d) 639

RRB NTPC 19.04.2016 Shift : 2

Ans : (a) Given series-
6 + 11 + 16 + 21 + ……. + 71
∵ This is an arithmetical progression in which-
a = 6, d = 5, l = 71
∴ l = a + (n –1) d
⇒ 71 = 6 + (n–1) 5
⇒ (n–1) = 13
⇒ n = 14

Hence, the required sum of the series $S_n = \frac{n}{2}(a + \ell)$

$$= \frac{14}{2}(6 + 71) = 7 \times 77 = \boxed{539}$$

37. If 11, 17, 23…….. be in an arithmetical progression. Find it's 12th term.

(a) 77 (b) 83
(c) 71 (d) 89

RRB NTPC 29.04.2016 Shift : 3

Ans. (a) : 11, 17, 23, ……..
a = 11
d = 17–11= 6
∴ T_n = a + (n –1).d
T_{12} = 11 + (12 – 1) 6
= 11 + 11 × 6 = 77

Type - 2

38. Which of the following numbers is prime?
(a) 263 (b) 243
(c) 253 (d) 273
RRB RPF Constable -17/01/2019 (Shift-III)

Ans : (a) Identifying the prime number
To identify a prime number, compare the given number with its nearest square. For example option (a), 263 (Which is between the squares of 16 and 17)

$16^2 = 256$

$17^2 = 289$

Then, divide all the prime numbers by 16 and 17. If the number is not divisible by any number then it is a prime number.
$\Rightarrow$ 263 (is less than the square of 17)
Which is not divisible by 2,3,5,7, 11 and 13.
Hence, it is a prime number.

39. Find the largest two-digit prime number.
(a) 93 (b) 89
(c) 91 (d) 97
RRB JE - 23/05/2019 (Shift-II)

Ans : (d) The number which is only 1 and divided itself is called prime number.
Hence, It is clear that the largest two digit prime number = 97

40. What will be the product of the smallest prime number (except 0) and any whole number?
(a) Always 0 (b) Always 1
(c) Always even number
(d) Always odd number
RRB RPF Constable -20/01/2019 (Shift-II)

Ans : (c) The smallest prime number = 2,
The result of the product of any whole number(except 0) and 2 is always an even number.

41. Find the sum of the prime numbers between 50 and 80.
(a) 392 (b) 390
(c) 463 (d) 396
RRB RPF Constable -18/01/2019 (Shift-I)

Ans : (c) Sum of prime numbers between 50 and 80 = 53 + 59 + 61 + 67 + 71 + 73 + 79 = 463

42. Which four odd prime numbers sum of 34?
(a) 1, 3, 5, 7 (b) 3, 5, 7, 9
(c) 3, 5, 11, 13 (d) 3, 7, 11, 13
RRB NTPC 04.04.2016 Shift : 2

Ans : (d) From option- (d)
$3+7+11+13=34$

43. How many times does the digit 2 come in place of tens in counting from 1 to 100?
(a) 20 (b) 11
(c) 10 (d) 19
RRB NTPC 31.03.2016 Shift : 1

Ans : (c) From 11 to 20 = 1 time
From 21 to 30 = 9 times
From 31 to 100 = 0 times
$\therefore$ Total number = 1 + 9 = 10

44. In a prime number……..
(a) There are more than two divisors.
(b) Divided by itself and 1.
(c) It has no divisor.
(d) Is not a positive integer.
RRB NTPC 30.03.2016 Shift : 2

Ans : (b) A prime number is divided by itself and 1 only.

45. How many times does the digit 5 come in the counting from 1 to 100?
(a) 21 (b) 22
(c) 20 (d) 19
RRB RPF SI-16/01/2019 (Shift-I)

Ans : (c) The total numbers in which 5 comes from 1 to 49 = 5
And from 50 to 60, such number = 11
And from 61 to 100, such number = 4
Hence, total required number = 5 + 11 + 4 = 20

46. How many total prime numbers are in first 200 odd natural numbers?
(a) 45 (b) 49
(c) 50 (d) 46
RRB Group 'D' 07/12/2018 (Shift-I)

Ans : (a) Total prime numbers in first 200 odd natural numbers = 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199, = 45

47. Which of the following pairs are co-primes?
(a) 348, 296 (b) 114, 213
(c) 59, 97 (d) 3025, 4920
RRB Group-D – 20/09/2018 (Shift-II)

Ans : (c) Such two numbers are called co-prime whose HCF is 1.
From option (c),
59, 97 is the appropriate co-prime pair in the alternatives.

48. Which of the following numbers is divisible?
(a) 719 (b) 709
(c) 729 (d) 739
RRB Group-D – 20/09/2018 (Shift-I)

Ans. (c) The number- 729 is divisible by 3, 9 and 81.

49. How many prime numbers are in first 100 natural numbers?
(a) 25 (b) 27
(c) 24 (d) 26
RRB Group-D – 26/11/2018 (Shift-III)

Ans : (a) Prime numbers in first 100 natural numbers = 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97
Therefore, total such numbers are 25.

50. Find out which of the following sets form co-prime numbers.
(a) (12, 7) (b) (21, 42)
(c) (3, 9) (d) (43, 129)
RRB NTPC 18.01.2017 Shift : 1

Ans : (a) Co-prime numbers- The set of two such numbers which HCF 1, is called co-prime numbers.
∴ In option (a) (12, 7), HCF of the numbers = 1

51. Which of the following is an odd composite number?
(a) 13 (b) 17
(c) 12 (d) 15
RRB NTPC 18.01.2017 Shift : 2

Ans : (d) The odd composite number will be 15.

52. Find the sum of first 8 odd prime numbers.
(a) 77 (b) 98
(c) 75 (d) 100
RRB NTPC 19.04.2016 Shift : 2

Ans : (b) First 8 odd prime numbers = 3, 5, 7, 11, 13, 17, 19, 23
Sum of the numbers = 3 + 5 + 7 + 11 + 13 + 17 + 19 + 23 = 98

53. How many prime numbers are between positive integers 60 and 100?
(a) 9 (b) 6
(c) 7 (d) 8
RRB NTPC 06.04.2016 Shift : 1

Ans : (d) The prime numbers between 60 and 100 = 61, 67, 71, 73, 79, 83, 89, 97
Hence, Total 8 prime numbers will be between 60 and 100.

54. Which of the following numbers is a prime number?
(a) 121 (b) 141
(c) 181 (d) 161
RRB ALP & Tec. (21-08-18 Shift-II)

Ans : (c) Prime numbers are divisible by 1 and itself only.
Example:- 5,11,13,19
From options,
Factors of the numbers,
181 = 1, 181
121 = 1, 11, 121
141 = 1, 3, 47, 141
161 = 1, 7, 23, 161

55. Which of the following pairs is NOT a pair of twin primes?
(a) 11, 13 (b) 71, 73
(c) 131, 133 (d) 191, 193
RRB ALP & Tec. (21-08-18 Shift-II)

Ans : (c) 131 and 133 does not have 133 prime number, because the factors of 1, 9, 7, 133.
From option (c),
Factors of 133 = 1, 7, 19, 133.
Hence, this pair is not a pair of twin primes.

56. Which of the following is the prime number series from 1 to 20?
(a) 3, 5, 7, 11, 13, 17, 19
(b) 2, 5, 7, 9, 11, 13, 17, 19
(c) 2, 3, 5, 7, 11, 13, 17, 19
(d) 1, 2, 3, 5, 7, 11, 13, 17, 19
RRB ALP & Tec. (20-08-18 Shift-I)

Ans : (c) Prime numbers:- The number divisible by 1 and itself only.
Hence, from options, the required series will be 2,3,5,7,11,13,17,19.

57. Which of the following number is not composite?
(a) 209 (b) 203
(c) 161 (d) 109
RRB ALP & Tec. (14-08-18 Shift-I)

Ans : (d) The number is called composite, which is formed by multiplying whole numbers.
Hence, 209 = 11 × 19
203 = 7 × 29
161 = 7 × 23
But, 109 cannot be expressed in the form of factors (except 1).

Type - 2

58. If x + y = 11, then $(-1)^x+(+1)^y$ is equal to ..... (where x and y are whole numbers).
(a) -1 (b) 1
(c) 2 (d) 0
RRB JE - 23/05/2019 (Shift-I)

Ans : (d) Given –
x + y = 11
$(-1)^x + (-1)^y = ?$
Note- When the sum of two whole numbers is an odd number then one will be even and second will be odd.
Hence,
$(-1)^{even/odd} + (-1)^{odd/even} = 0$

59. From a cloth of 30 m long, 12 pieces each measuring 225 cm are cut and sold. How much is left of the original length?
(a) 1/3 (b) 1/9
(c) 1/10 (d) 3/10
RRB JE - 23/05/2019 (Shift-III)

Ans : (c) The total length of the cloth = 30 m [1m = 100 cm] = 3000 cm
Total length of the cloth that is sold = 225 × 12 = 2700 cm
The length of the remaining cloth = 3000 – 2700 = 300 cm
The remaining part = $\frac{300}{3000} = \frac{1}{10}$ fraction

60. If 1/7 of a number is subtracted from the number, the result is 30 less than the number. Find the number.
(a) 105 (b) 140

(c) 120 (d) 210

RRB JE - 24/05/2019 (Shift-III)

Ans : (d) Let the number be x.

$$x - \frac{x}{7} = x - 30$$

$$\frac{7x - x}{7} = x - 30$$

$6x = 7x - 210$

$x = 210$

61. If the product of two numbers is 24, and their square's sum is 52, then find their sum.

(a) 5 (b) 10

(c) 15 (d) 20

RRB RPF Constable -24/01/2019 (Shift-I)

Ans :(b) Let the numbers be x and y.

According to the question,

$x.\ y = 24$……..(1)

$x^2 + y^2 = 52$……(2)

$\because (x + y)^2 = x^2 + y^2 + 2xy$

$= 52 + 2\times24$

$= 52 + 48 = 100$

$x + y = \sqrt{100} = 10$

62. If 10 is subtracted from the 5 times of a number, then that number will be equal to the number found when adding 8 to 4 times that number, what is that number?

(a) 15 (b) 18

(c) 22 (d) 21

RRB RPF Constable -25/01/2019 (Shift-III)

Ans. (b) : Let the number be x.

According to the question,

$5x - 10 = 4x + 8$

$x = 18$

Hence, the required number will be 18.

63. When 8 times of a number is added to 4, the result obtained is the smallest 3-digit number. What is that number?

(a) 12 (b) 10

(c) 15 (d) 8

RRB RPF Constable -22/01/2019 (Shift-II)

Ans : (a) Let the number be x,

$\because$ The smallest 3-digit number = 100

According to the question,

$8x + 4 = 100$

$8x = 96$

$$x = \frac{96}{8} = 12$$

Hence, the required number will be 12.

64. The sum of two numbers is 22. Five times of one number is equal to 6 times the other. Find the larger of the two numbers.

(a) 12 (b) 15

(c) 10 (d) 16

RRB JE - 25/05/2019 (Shift-I)

Ans : (a) Let the numbers are x and y,

According to the question

$x + y = 22$ .....(i)

and $5x = 6y$ .....(ii)

$$x = \frac{6}{5}y$$

Putting the value of x in equation (i) -

$$\frac{6}{5}y + y = 22$$

$$\frac{11}{5}y = 22$$

$$y = \frac{22\times5}{11} = 10$$

$y = 10$

$$\therefore\ x = \frac{6}{5}\times10 = 12$$

Hence, the larger number is 12.

65. If doubling a number and adding 20 to the result gives the same answer as multiplying the number by 8 and subtracting 4 from the product, find the number.

(a) 3 (b) 4

(c) 6 (d) 2

RRB JE - 25/05/2019 (Shift-II)

Ans : (b) Let the number be = x

According to the question,

$2x + 20 = x \times 8 - 4$

$2x + 20 = 8x - 4$

$24 = 6x$

$\boxed{x = 4}$

66. The product of two numbers is 9375. The quotient, when the largest number is divided by the smallest number the quotient is 15. Find the sum of these numbers.

(a) 400 (b) 380

(c) 425 (d) 395

RRB JE - 30/05/2019 (Shift-II)

Ans : (a) Let the smaller number be = x

$\therefore$ Larger number = 15x

According to the question,

$x \times 15x = 9375$

$15x^2 = 9375$

$x^2 = 625$

x = 25 first number

$\therefore$ 15x = 15 × 25 = 375 second number

Hence, the sum of the numbers = 375 + 25 = 400

67. If 2/3rd of 1/4th of a number is 32. Find the number.

(a) 202 (b) 198

(c) 196 (d) 192

RRB RPF-SI -13/01/2019 (Shift-III)

Ans : (d) Let the number be x,

$$x \times \frac{1}{4} \times \frac{2}{3} = 32$$

$$x = 32 \times 6 = 192$$

68. If the sum of two numbers is 13 and the sum of their squares is 97, what is their product?

(a) 72 (b) 36
(c) 110 (d) 84

RRB JE - 28/06/2019 (Shift-III)

Ans. (b) Let both numbers are X and Y.
Given,

$x + y = 13, \quad x^2 + y^2 = 97, \quad xy = ?$

$\because (x + y)^2 = x^2 + y^2 + 2xy$ ..... (i)

Putting the value in equation (i),

$(13)^2 = 97 + 2xy$

$169 = 97 + 2xy$

$2xy = 169 - 97$

$xy = \frac{72}{2}$

$xy = 36$

69. Which of the fractions given below, when added to $\frac{13}{5}$, gives 1?

(a) $-\frac{48}{30}$ (b) $-\frac{7}{5}$
(c) $-\frac{28}{10}$ (d) $-\frac{8}{15}$

RRB Group-D – 19/09/2018 (Shift-II)

Ans. (a) : Let the unknown fraction be x.
According to the question,

$$x + \frac{13}{5} = 1$$

$$x = 1 - \frac{13}{5}$$

$$x = \frac{-8}{5}$$

or , $x = \frac{-8 \times 6}{5 \times 6} = \frac{-48}{30}$

70. Find the answer of $\frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \frac{1}{56}$

(a) $\frac{11}{24}$ (b) $\frac{3}{8}$
(c) $\frac{7}{16}$ (d) $\frac{13}{28}$

RRB Group-D – 03/10/2018 (Shift-II)

Ans : (b) From the question,

$$\frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \frac{1}{56}$$

$$= \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \frac{1}{4} - \frac{1}{5} + \frac{1}{5} - \frac{1}{6} + \frac{1}{6} - \frac{1}{7} + \frac{1}{7} - \frac{1}{8}$$

$$= \frac{1}{2} - \frac{1}{8} = \frac{4-1}{8} = \frac{3}{8}$$

71. Shalini, Tanvir and Rashid shared a cake. Shalini had $\frac{1}{6}$ part of it, Tanvir had $\frac{1}{4}$ part of it and Rashid had the remaining part. What was fraction of Rashid's cake?

(a) $\frac{5}{6}$ (b) $\frac{3}{5}$
(c) $\frac{13}{15}$ (d) $\frac{7}{12}$

RRB Group-D – 31/10/2018 (Shift-II)

Ans : (d) Shalini's share of the cake $= \frac{1}{6}$ part

Tanvir's share of the cake $= \frac{1}{4}$ part

Total share of Shalini and Tanvir's cake

$$= \frac{1}{6} + \frac{1}{4} = \frac{2+3}{12} = \frac{5}{12}$$

Hence, Rashid's share of the cake $= 1 - \frac{5}{12} = \boxed{\frac{7}{12}}$ part

72. The sum of two numbers is 9. The sum of their reciprocals is 1/2. One of the number is.

(a) 2 (b) 4
(c) 5 (d) 6

RRB Group-D – 17/09/2018 (Shift-III)

Ans. (d) : Let the first number be x and the second number be y.
According to the question,

$x + y = 9$ .....(i)

$\frac{1}{x} + \frac{1}{y} = \frac{1}{2}$ ..... (ii)

From equation (i),

$x + y = 9$

$y = 9 - x$

From equation (ii)

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{2}$$

$$\frac{x+y}{xy} = \frac{1}{2}$$

$$\frac{9 \times 2}{xy} = 1$$

$2 \times 9 = xy$

Putting the value of y,

$18 = x(9 - x)$

$18 = 9x - x^2$

$x^2 - 9x + 18 = 0$

$x^2 - 6x - 3x + 18 = 0$

$x(x - 6) - 3(x - 6) = 0$

$(x - 3)(x - 6) = 0$

$(x - 3) = 0$ or $x = 3$

$(x - 6) = 0$ or $x = 6$

73. There are a total of 200 students in a school, of which $\frac{7}{10}$ are boys. Find the number of girls in the school.

(a) 80 (b) 60
(c) 40 (d) 120

RRB NTPC 17.01.2017 Shift-3

Ans : (b) The number of boys in the school

$= \text{total students} \times \frac{7}{10}$

$= 200 \times \frac{7}{10} = 140$

$\therefore$ The number of girls in the school = 200–140 = 60

74. If the sum of two numbers is 26 and their difference is 12. Find the difference of their squares.

(a) 296 (b) 312
(c) 324 (d) 336

RRB NTPC 05.04.2016 Shift : 2

Ans : (b) Let the numbers be x and y.
x + y = 26
x – y = 12
$\therefore$ The difference of the squares,
$= x^2 - y^2$
$= (x + y)(x - y)$
$= 26 \times 12 = 312$

75. If the product of two numbers is thrice of their sum, if 1st number is 12 find the 2nd number.

(a) 2 (b) 3
(c) 4 (d) 5

RRB NTPC 04.04.2016 Shift : 1

Ans : (c) Let the 2nd number be x.
$x \times 12 = (x + 12) \times 3$
$12x = 3x + 36$
$9x = 36$
Hence, $x = 4$

76. If the sum of two digits is 15 and difference is 3. The digits are-

(a) 8 and 7 (b) 6 and 9
(c) 5 and 10 (d) 3 and 12

RRB NTPC 31.03.2016 Shift : 1

Ans : (b) Let the digits be x and y.
Then, $x + y = 15$ …… (1)
$x - y = 3$ ……. (2)
Solving equation (1) and (2),
$x = 9$
$y = 6$

77. The sum of three consecutive even numbers is 42. Find the middle number.

(a) 12 (b) 18
(c) 16 (d) 14

RRB NTPC 29.03.2016 Shift : 3

Ans : (d) Let three consecutive even numbers are x, x + 2 and x+4
$\therefore x + x + 2 + x + 4 = 42$
$3x = 36$
$x = 12$
$\therefore$ middle number = 12+2=14

78. Two partners M and N buy a car. M pays his share of $\frac{3}{7}$th of the total cost of the car. M pays ₹31,540 as compared to N. What is the cost of the car?

(a) ₹2,32,680 (b) ₹2,03,175
(c) ₹2,20,780 (d) ₹1,85,780

RRB ALP & Tec. (31-08-18 Shift-III)

Ans : (c) Let the cost of the car is x rupee
According to the question,

M's share $= \frac{3x}{7}$

N's share $= \frac{3x}{7} + 31540$

Then, $\frac{3x}{7} + 31540 + \frac{3x}{7} = x$
$x = 31540 \times 7$
x = ₹2,20,780

79. If $\frac{2}{3}$ part of a pizza costs Rs. 300, then $\frac{3}{5}$ part of a pizza will cost:

(a) ₹180 (b) ₹250
(c) ₹225 (d) ₹270

RRB ALP & Tec. (30-08-18 Shift-I)

Ans : (d) The cost of 2/3 part of the pizza = ₹300

Then, the cost of 1 share of the pizza $= \frac{300 \times 3}{2}$ = ₹450

The cost of 3/5th share of the pizza $= 450 \times \frac{3}{5}$
$= 90 \times 3$ = ₹270

80. When 472 pieces of plywood, each 0.23 cm thick, are placed on top of each other, what would be the height of the pillar in metre?

(a) 10.856 (b) 1.0856
(c) 108.56 (d) 1.856

RRB ALP & Tec. (29-08-18 Shift-III)

Ans : (b) The required height of the pillar,

$= \frac{0.23 \times 472}{100}$ = 1.0856 metre

81. 15 small rods, each of length $23\frac{2}{7}$m are joined to make a big rod. What then is the length of the big rod?

(a) $349\frac{3}{7}$m (b) $349\frac{1}{7}$m
(c) $349\frac{2}{7}$m (d) $349\frac{5}{7}$m

RRB ALP & Tec. (21-08-18 Shift-I)

Ans : (c) The length of each rod $= 23\frac{2}{7} = \frac{163}{7}$m

In this way, the length of all 15 rods $= 15 \times \frac{163}{7}$

$= \frac{2445}{7} = 349\frac{2}{7}$m

Type - 2

82. Calculate the difference between the largest and the smallest two-digit prime number.

(a) 82 (b) 83
(c) 84 (d) 86

RRB RPF Constable -17/01/2019 (Shift-I)

Ans : (d) The largest two digit prime number = 97
The smallest two digit prime number = 11
Hence, the required number = 97 – 11 = 86

83. Find the largest number of four digit that is completely divisible by 49.

(a) 9998 (b) 9994
(c) 9992 (d) 9996

RRB RPF-SI -10/01/2019 (Shift-II)

Ans : (d) The largest 4-digit number is 9999.

```
49)9999(204
   98
   199
   196
     3
```

Hence, the required number = 9999-3 = 9996, which is exactly divisible by 49.

84. Find the greatest 4-digit number that is divisible by 15, 25, 40 and 75.

(a) 9200 (b) 9600
(c) 9400 (d) 9000

RRB JE - 24/05/2019 (Shift-I)

Ans : (b) The largest 4-digit number = 9999
LCM of 15, 25, 40 and 75 = 600

```
600)9999(16
    600
    3999
    3600
     399
```

Hence, the largest 4-digit number that is divisible by 15, 25, 40 and 75 = 9999 – 399 = 9600

85. What should be added to 135642 to get the largest six digit number?

(a) 864350 (b) 863357
(c) 864357 (d) 864347

RRB Group-D – 29/10/2018 (Shift-III)

Ans : (c) From option, Let the required number is x.
Adding x to 135642 gives a largest six digit number
∴ 135642 + x = 999999
x = 999999 – 135642
x = 864357

86. Find the largest 4 digit number that is completely divisible by 88.

(a) 9944 (b) 9844
(c) 9768 (d) 8894

RRB JE - 02/06/2019 (Shift-I)

Ans : (a) The largest 4 digit number = 9999

```
      113
88)9999
   88
   119
   88
   319
   264
    55   Remainder
```

Hence, the required number = 9999 – 55 = 9944

86. Find the largest number of four digits that is completely divisible by 27, 18, 15 and 12.

(a) 9730 (b) 9710
(c) 9700 (d) 9720

RRB JE - 26/06/2019 (Shift-III)

Ans : (d) LCM of 12, 15, 18 and 27 = 540

```
540)  9999  (18
      540
      4599
      4320
       279
```

Hence, the required number = 9999 – 279 = 9720

88. What is the largest four-digit number that is completely divisible by 49?

(a) 9992 (b) 9994
(c) 9998 (d) 9996

RRB Group-D – 18/09/2018 (Shift-II)

Ans. (d) : The largest 4 digit number = 9999

```
49)9999(204
   98
   199
   196
   ××3
```

Hence, the largest 4 digit number completely divisible by 4 = 9999 – remainder = 9999 – 3 = 9996

89. Find the smallest four digit number divisible by 47.

(a) 1200 (b) 1025
(c) 1034 (d) 1360

RRB Group-D – 22/09/2018 (Shift-III)

Ans. (c) : The smallest four digit number = 1000

```
      21
47)1000
   94
   60
   47
   13
```

Hence, the smallest four digit number divisible by 47,
= 1000 + (47–13) 1000+34 = 1034

90. Find the largest 3-digit number that is completely divisible by 10, 8 and 12.

(a) 940 (b) 960
(c) 980 (d) 999

RRB Group-D – 26/09/2018 (Shift-III)

Ans : (b) LCM of 10, 12 and 8

2	10,	8,	12
2	5,	4,	6
2	5,	2,	3
3	5,	1,	3
5	5,	1,	1
	1,	1,	1

LCM = 2 ×2×2×3×5 = 120
The largest 3-digit number = 999
According to the question-

$$120\overline{)999}\ (8,\ 960,\ 39)$$

Hence, the required number = 999 – 39 = 960.

91. Find the smallest four digit number which is a perfect square.

(a) 1000 (b) 1024
(c) 1081 (d) 1064

RRB NTPC 04.04.2016 Shift : 1

Ans : (b) The smallest 4 digit number = 1000

	32
3	1000
+3	9
62	100
2	124
	–24

Hence, the smallest 4 digit number = 1000 + 24 = 1024

92. Find the least 6 digit number that is a multiple of 18.

(a) 100000 (b) 999900
(c) 100008 (d) 100006

RRB NTPC 29.04.2016 Shift : 1

Ans : (c) The smallest 6 digit number = 100000

$$18\overline{)100000}\ (5555,\ 90,\ 100,\ 90,\ 100,\ 90,\ 100,\ 90,\ 10)$$

The remainder is 10, hence 8 is added to the number will make it divisible totally.
Hence, the required number = 100000 + 8 = 100008

Type - 6

93. Which from the following is not a rational number?

(a) $\sqrt[3]{1728}$ (b) π
(c) 2.487627287 (d) 8.36712846781

RRB RPF-SI -05/01/2019 (Shift-I)

Ans : (b) Irrational numbers is a real number cannot be expressed as p/q. where p and q are integers and q is not 0. It means, irrational number cannot be expressed as fractions. From example the square root of 2 π are irrational number.

94. Which from the following is not irrational?

(a) $\sqrt{5428}$ (b) $\sqrt{6084}$
(c) π (d) $\sqrt{7652}$

RRB RPF Constable -18/01/2019 (Shift-III)

Ans. (b) : The real numbers which cannot be expressed as p/q, where p and q are integers and q is not 0, are called irrational numbers. These numbers are represented by Q^C or Q^I.
For example- $\sqrt{2}, 1+\sqrt{3}, \pi$
$\sqrt{6084} = \sqrt{78\times 78} = 78$ (Rational number)

95. Denote $\overline{0.125}$ as a rational number.

(a) 119/993 (b) 113/990
(c) 125/999 (d) 100/999

RRB JE - 25/05/2019 (Shift-I)

Ans : (c) Let x = $0.\overline{125}$
x = 0.125125 ..... (i)
1000x = 125.125125 ..... (ii)
From equation (ii)- equation (i) –
999x = (125.125125.....) – (0.125125.....)
999x = 125.0
$x = \frac{125}{999}$

96. Denote 5.025 as a fraction:

(a) $\frac{203}{40}$ (b) $5\frac{1}{40}$
(c) $5\frac{1}{4}$ (d) $5\frac{1}{8}$

RRB JE - 29/05/2019 (Shift-I)

Ans : (b) 5.025 $= \frac{5025}{1000}$
$= \frac{201}{40}, = 5\frac{1}{40}$

97. Find the smallest fraction.

(a) 3/4 (b) 11/13
(c) 5/7 (d) 9/11

RRB JE - 27/06/2019 (Shift-III)

Ans : (c) (a) 3/4 = 0.75
(b) 11/13 = 0.84
(c) 5/7 = 0.71
(d) 9/11 = 0.81
Hence, the smallest fraction will be 5/7.

98. Find the value of the denominator of $\frac{1}{(5+\sqrt{3})}$ in rational number.

(a) $\frac{(5-\sqrt{3})}{22}$ (b) $5+\frac{\sqrt{3}}{22}$

(c) $5-\frac{\sqrt{3}}{20}$ (d) $\frac{(5-\sqrt{3})}{20}$

RRB Group-D – 29/10/2018 (Shift-III)

Ans : (a)

According to the question-

$$\frac{1}{(5+\sqrt{3})}=\frac{(5-\sqrt{3})}{(5+\sqrt{3})(5-\sqrt{3})}$$

$$=\frac{(5-\sqrt{3})}{(5)^2-(\sqrt{3})^2}$$

$$=\frac{(5-\sqrt{3})}{25-3} \qquad =\frac{(5-\sqrt{3})}{22}$$

99. Which of the following square roots is irrational?

(a) 21025 (b) 18025

(c) 13225 (d) 15625

RRB Paramedical Exam – 21/07/2018 (Shift-I)

Ans. (b) : $\sqrt{21025}=\sqrt{5\times5\times29\times29}$

= 145 (Rational number)

$\sqrt{18025}=\sqrt{5\times5\times7\times103}$

= 135.257 (Irrational number)

$\sqrt{13225}=\sqrt{5\times5\times23\times23}$

= 5×23=115 (Rational number)

$\sqrt{15625}=\sqrt{5\times5\times5\times5\times5\times5}$

= 5×5×5=125(Rational number)

Hence, it is obvious that the square root of 18025 is irrational number.

100. Find the rational value of the denominator of 1/(2+$\sqrt{3}$).

(a) $2+\sqrt{3}$ (b) $2-\sqrt{3}$

(c) 1 (d) $4+\sqrt{3}$

RRB Group-D – 22/10/2018 (Shift-III)

Ans : (b) Rationalizing the denominator of $\frac{1}{2+\sqrt{3}}$,

$$\frac{1}{2+\sqrt{3}}\times\frac{2-\sqrt{3}}{2-\sqrt{3}}$$

$$=\frac{2-\sqrt{3}}{2^2-(\sqrt{3})^2}$$

$$=\frac{2-\sqrt{3}}{4-3}=2-\sqrt{3}$$

101. Find the rational value of the denominator of $\frac{1}{(5+2\sqrt{3})}$

(a) $\frac{(5-2\sqrt{3})}{12}$ (b) $\frac{(5-2\sqrt{3})}{13}$

(c) $5-\frac{2\sqrt{3}}{13}$ (d) $5+\frac{2\sqrt{3}}{13}$

RRB Group-D – 25/10/2018 (Shift-II)

Ans : (b) Rationalizing the denominator of the given fraction,

$$=\frac{1}{(5+2\sqrt{3})}\times\frac{(5-2\sqrt{3})}{(5-2\sqrt{3})}$$

$$=\frac{(5-2\sqrt{3})}{(5)^2-(2\sqrt{3})^2} \qquad [(a+b)(a-b) = a^2-b^2]$$

$$=\frac{5-2\sqrt{3}}{25-12}=\frac{5-2\sqrt{3}}{13}$$

102. $(4)^{\frac{-3}{2}}$ = ?

(a) 1/4 (b) 8

(c) 1/8 (d) 4

RRB NTPC 07.04.2016 Shift : 3

Ans :(c) $(4)^{\frac{-3}{2}}=(2)^{2\times\frac{-3}{2}}$

$$=2^{-3}=\frac{1}{2^3}=\frac{1}{8}$$

103. From the given options, find the rational number between the range 2/4 and 0.6.

(a) $\frac{11}{25}$ (b) $\frac{21}{40}$

(c) $\frac{3}{4}$ (d) $\frac{11}{4}$

RRB NTPC 19.01.2017 Shift : 2

Ans : (b) The rational number between $\frac{2}{4}=0.5$ and 0.6

$=\frac{21}{40}=0.525$

Hence, 0.5< 0.525< 0.6

104. Which of the following numbers is irrational?

(a) $\sqrt[3]{64}$ (b) $\sqrt{64}$

(c) $\sqrt[6]{64}$ (d) $\sqrt[4]{64}$

RRB ALP & Tec. (30-08-18 Shift-I)

Ans : (d) From options,

$\sqrt[3]{64}=(64)^{\frac{1}{3}}=(4^3)^{\frac{1}{3}}=4$ (Rational number)

$\sqrt{64}=(64)^{\frac{1}{2}}=(8^2)^{\frac{1}{2}}=8$ (Rational number)

$\sqrt[6]{64}=(64)^{\frac{1}{6}}=(2^6)^{\frac{1}{6}}=2$ (Rational number)

$\sqrt[4]{64}=\sqrt[4]{16}\times\sqrt[4]{4}=2\times\sqrt[4]{4}=$ (Irrational number)

105. Among the following which is a rational number?

(a) $\sqrt[3]{2}$ (b) $\sqrt[3]{8}$

(c) $\sqrt[3]{4}$ (d) $\sqrt[3]{12}$

RRB ALP & Tec. (13-08-18 Shift-III)

Ans : (b) Rational number can be written as p/q ;(q $\neq$ 0).
From option (b), is rational number

$$\sqrt[3]{8} = 2$$

106. Which of the numbers given below is NOT rational number?

(a) $\sqrt{64}$ (b) $\sqrt[3]{64}$

(c) $\sqrt[3]{8}$ (d) $\sqrt{8}$

RRB ALP & Tec. (09-08-18 Shift-II)

Ans : (d) $\sqrt{64} = 8$ (Rational number)

$\sqrt[3]{64} = 4$ (Rational number)

$\sqrt[3]{8} = 2$ (Rational number)

$\sqrt{8} = 2\sqrt{2}$ (Irrational number)

Hence $2\sqrt{2}$ is not number a rational

A number which we can write as p/q where p and q both are integers but q $\neq$ 0 is called rational numbers.

107. All irrational number are---------numbers.

(a) Integers (b) Imaginary

(c) Whole (d) Real

RRB NTPC 19.01.2017 Shift : 3

Ans : (d) All irrational numbers are real numbers.
As- $\sqrt{2}$

108. Which from the following is irrational?

(a) $\sqrt{1000000}$ (b) $\sqrt[3]{1000000}$

(c) $\sqrt[6]{1000000}$ (d) $\sqrt[4]{1000000}$

RRB Group-D – 08/10/2018 (Shift-II)

Ans : (d) From option-

(a) $\sqrt{1000000}$

$\sqrt{100\times100\times100} = 10\times10\times10 = 1000$ (Rational)

(b) $\sqrt[3]{1000000} = \left(100^3\right)^{1/3} = 100$ (Rational)

(c) $\sqrt[6]{1000000} = \left(10^6\right)^{1/6} = 10$ (Rational)

(d) $\sqrt[4]{1000000} = 10\sqrt[4]{100}$ (Irrational)

109. Which from the following is an irrational number?

(a) $\sqrt[4]{4}$ (b) $\sqrt[3]{8}$

(c) $\sqrt{16}$ (d) $\sqrt[6]{1}$

RRB Group-D – 22/09/2018 (Shift-III)

Ans. (a) : Irrational number – The number which cannot be expressed as p/q. Example- $\sqrt{3}, \sqrt{6}.....\sqrt[4]{4}$

Note- π is an irrational number while 22/7 is a rational number.
According to the question,

$\sqrt[3]{8} = \sqrt[3]{2\times2\times2} = 2$ (Rational number)

$\sqrt{16} = \sqrt{2\times2\times2\times2} = 2\times2 = 4$ (Rational number)

$\sqrt[6]{1} = 1$ (Rational number)

110. Which from the following is a rational number?

(a) $\sqrt[3]{2} - 2$ (b) $\sqrt[3]{8} - 2$

(c) $\sqrt[3]{4} + 4$ (d) $\sqrt[3]{12} + 1$

RRB Group-D – 25/09/2018 (Shift-I)

Ans : (b)
Rational number – Such numbers which can be expressed as p/q ;(q $\neq$ 0), are called rational numbers.
Example:- $\sqrt[3]{8}, \sqrt{4}$ etc;
Irrational number – Such numbers which cannot be expressed as p/q. Example:- $\pi, \sqrt[3]{2}, \sqrt{2}$ etc;
From options,
(b) $\sqrt[3]{8} - 2 = 0$ is a rational number while others $\sqrt[3]{2} - 2$, $\sqrt[3]{4} + 4$ are irrational numbers.

111. Which from the following is an irrational number?

(a) $\sqrt[4]{1024}$ (b) $\sqrt[10]{1024}$

(c) $\sqrt{1024}$ (d) $\sqrt[5]{1024}$

RRB Group-D – 27/09/2018 (Shift-I)

Ans. (a) From option,

(a) $\sqrt[4]{1024} = 4\sqrt[4]{4}$ (Irrational number)

(b) $\sqrt[10]{1024} = 2$ (Rational number)

(c) $\sqrt{1024} = 32$ (Rational number)

(d) $\sqrt[5]{1024} = 4$ (Rational number)

112. Which from the following is not a rational number?

(a) $\sqrt[5]{32}$ (b) $\sqrt[3]{64}$

(c) $\sqrt[4]{32}$ (d) $\sqrt[3]{27}$

RRB Group-D – 28/09/2018 (Shift-I)

Ans : (c) from options,

$\sqrt[5]{32} =$ 2(Rational)

$\sqrt[3]{64} = 4$ (Rational)

$\sqrt[3]{27} = 3$ (Rational)

$\sqrt[4]{32} = 2\sqrt[4]{2}$ (Irrational)

113. Which from the following is a rational number?

(a) $\sqrt[5]{1551}$ (b) $\sqrt[3]{1331}$

(c) $\sqrt{1221}$ (d) $\sqrt[4]{1441}$

RRB Group-D – 11/10/2018 (Shift-III)

Ans : (b) A rational number can be written as p/q ;(q $\neq$ 0).
Hence, $\sqrt[3]{1331} = \sqrt[3]{11\times11\times11} = 11$
Therefore, 11/1 is a rational number.
Note- All other numbers cannot be written as p/q. So they are irrational.

114. Whose square root from the following numbers is a rational number?

(a) 576 (b) 512
(c) 480 (d) 544

RRB Group-D – 07/12/2018 (Shift-III)

Ans : (a) From the options, the square root of 576 is rational.

$\sqrt{576} = 24$
$\sqrt{512} = 16\sqrt{2}$
$\sqrt{480} = 4\sqrt{30}$
$\sqrt{544} = 4\sqrt{34}$

Hence, square root of 576 is 24, which is a rational number.

115. Whose square root from the following numbers will be rational?

(a) 46232 (b) 46233
(c) 14448 (d) 34225

RRB Group-D – 06/12/2018 (Shift-II)

Ans. (d) : The square root of 34225 is 185, which is a rational number.

116. Whose square root from the following numbers is irrational?

(a) 5184 (b) 4465
(c) 3025 (d) 8836

RRB Group-D – 05/12/2018 (Shift-II)

Ans. (b) From options,

(a) $\sqrt{5184} = 72$
(b) $\sqrt{4465} = \sqrt{5 \times 19 \times 47}$ (Irrational)
(c) $\sqrt{3025} = 55$
(d) $\sqrt{8836} = 94$

Hence, the square root of 4465 is irrational.

117. Whose square root from the following numbers is rational?

(a) 336 (b) 344
(c) 320 (d) 324

RRB Group-D – 04/12/2018 (Shift-III)

Ans. (d) $\sqrt{324} = \sqrt{18 \times 18} = 18$

Hence, the square root of 324 will be 18, which is a rational number.

118. Whose square root from the following numbers will be irrational?

(a) 6441 (b) 9604
(c) 7921 (d) 5776

RRB Group-D – 03/12/2018 (Shift-II)

Ans : (a) From option,

$\sqrt{6441} = 80.25$ is irrational
$\sqrt{9604} = \sqrt{98 \times 98} = 98$
$\sqrt{7921} = \sqrt{89 \times 89} = 89$
$\sqrt{5776} = \sqrt{76 \times 76} = 76$

Hence, the square root of 6441 will be irrational.

119. Whose square root from the following numbers is a rational number?

(a) 144 (b) 136
(c) 128 (d) 120

RRB Group-D – 01/12/2018 (Shift-II)

Ans : (a)

$\sqrt{144} = 12$
$\sqrt{136} = 2\sqrt{34}$
$\sqrt{128} = 8\sqrt{2}$
$\sqrt{120} = 2\sqrt{30}$

Hence, the square root of 144 is 12, that is a rational number.

120. Express $\frac{1}{(2+\sqrt{3})}$ as a rational number.

(a) $5-2\sqrt{3}/12$ (b) $(2-\sqrt{3})/1$
(c) $(5-2\sqrt{3})/13$ (d) $5+2\sqrt{3}/13$

RRB Group-D – 08/10/2018 (Shift-I)

Ans. (b) : Rationalizing the number $\frac{1}{(2+\sqrt{3})}$

$$= \frac{1 \times (2-\sqrt{3})}{(2+\sqrt{3})(2-\sqrt{3})}$$

$$= \frac{(2-\sqrt{3})}{(4-3)} = \frac{(2-\sqrt{3})}{1}$$

Type - 2

121. Find the least number that the sum obtained by to be adding in 1780 to is a perfect square.

(a) 46 (b) 49
(c) 69 (d) 72

RRB JE - 27/05/2019 (Shift-II)

Ans : (c) The number 1780 when added to 69 will be 1849 which is a perfect square number.

Thus-

1780 + 69 = 1849
1849 = 43 × 43
$(43)^2 = 1849$

122. Find the smallest integer whose cube is equal to itself.

(a) -1 (b) 2
(c) 1 (d) 0

RRB JE - 22/05/2019 (Shift-I)

Ans : (a) -1 and 1 are such integers whose cube is equal to itself.

Hence, the smallest integer = -1

$\because (-1)^3 = -1$

123. If the cube of a number and subtracting from $(153)^2$ to the result gives 1457. Find the number.

(a) 18 (b) 16
(c) 28 (d) 24

RRB JE - 24/05/2019 (Shift-I)

Ans : (c) Let the number be x.
According to the question,

$(153)^2 - x^3 = 1457$
$x^3 = (153)^2 - 1457$
$x^3 = 23409 - 1457$
$x^3 = 21952$
$\therefore\ x = \sqrt[3]{21952} = \sqrt[3]{28\times28\times28} = 28$

124. Five times of a positive integer is 3 less than twice of its square. Find the integer.
(a) 3 (b) 8
(c) 2 (d) 5

RRB RPF Constable -19/01/2019 (Shift-I)

Ans : (a) Let the positive integer is x.
According to the question-

$5x = 2x^2 - 3$
$2x^2 - 5x - 3 = 0$
$2x^2 - 6x + x - 3 = 0$
$2x(x-3) + 1(x-3) = 0$
$(x-3)(2x+1) = 0$
$x - 3 = 0\ 2x + 1 = 0$
$x = 3$ या $x = -½$ (Invalid)

125. Find the least perfect square that is divisible by 21, 36 and 66.
(a) 214344 (b) 231444
(c) 214434 (d) 213444

RRB JE - 25/05/2019 (Shift-III)

Ans : (d) LCM of 21, 36 and 66 –

2	21	36	66
2	21	18	33
3	21	9	33
3	7	3	11
7	7	1	11
11	1	1	11
	1	1	1

Hence, To get the required number, multiply by 7 and 11 in LCM of 21, 36, and 66.
The required perfect square number
$= 2\times2\times3\times3\times7\times7\times11\times11 = 213444$

126. Which of these square numbers cannot be expressed as the sum of two prime numbers?
(a) 81 (b) 49
(c) 121 (d) 144

RRB JE - 30/05/2019 (Shift-II)

Ans : (c) $\sqrt{81} = 9 = 2+7$
$\sqrt{49} = 7 = 2+5$
$\sqrt{121} = 11 = 2+9,\ 3+8,\ 5+6,\ 7+4$
$\sqrt{144} = 12 = 5+7$
Hence, Option (c) cannot be expressed as the sum of two prime numbers.

127. Three times the square of a number subtracting by 4 times the number is equal to 50 more than the number. Find the number.
(a) 5 (b) 4
(c) 6 (d) 10

RRB JE - 28/05/2019 (Shift-II)

Ans : (a) Let the number be = x
According to the question,

$3x^2 - x \times 4 = x + 50$
$3x^2 - 4x - x - 50 = 0$
$3x^2 - 5x - 50 = 0$
$3x^2 - 15x + 10x - 50 = 0$
$3x(x-5) + 10(x-5) = 0$
$(x-5)(3x+10) = 0$
$x - 5 = 0$
$x = 5$

128. Which of the following is not a perfect square?
(a) 2025 (b) 16641
(c) 1250 (d) 9801

RRB RPF Constable -20/01/2019 (Shift-I)

Ans : (c) From options- 1250 is not a perfect square, while others are perfect squares.
$1250 = (35.36)^2$ is not a perfect square
$2025 = (45)^2$
$16641 = (129)^2$
$9801 = (99)^2$

129. Which of these numbers is not a sum of two squares?
(a) 41 (b) 13
(c) 23 (d) 37

RRB JE - 26/06/2019 (Shift-I)

Ans : (c) From option-
(a) $41 = 5^2 + 4^2$ (b) $13 = 2^2 + 3^2$
(c) 23 (d) $37 = 6^2 + 1^2$
Hence the number 23 is not the sum of two square.

130. Which of these is a perfect square?
(a) 9801 (b) 9887
(c) 9013 (d) 9016

RRB JE - 01/06/2019 (Shift-III)

Ans. (a) From option (a),

	99
9	98 01
9	81
189	17 01
9	17 01
	××××

Hence, 9801 is a perfect square of 99.

131. If the last digit of the square of a number is 1. Find the last digit of it's cube.
(a) Only 9 (b) 1 or 9
(c) Any odd number (d) Only 1

RRB JE - 27/06/2019 (Shift-I)

Ans : (b) Let the number be 9. The last digit of whose square is 1. Which is as follows-

$9^2 = 81$

Last digit of 729 which is cube of 9 = 9

Let the number be 11. The last digit of whose square is 1. Which is as follows-

$11^2 = 121$

The last digit of the cube of 11-

$11^3 = 1331$

Hence the last digit = 1

Hence the number will be 1 or 9.

132. The sum and the difference of two numbers are 25 and 3 respectively. Find the difference of their squares.

(a) 165 (b) 75
(c) 154 (d) 140

RRB JE - 27/06/2019 (Shift-III)

Ans : (b) Let the two numbers are x and y

$x + y = 25$ ----(i)

$x - y = 3$ ----(ii)

$x^2 - y^2 = (x + y)(x-y) = 25 \times 3 = 75$

Hence, the difference of their squares = 75

133. How many perfect squares are there between 100 and 200?

(a) 7 (b) 4
(c) 6 (d) 5

RRB JE - 27/06/2019 (Shift-III)

Ans : (b) Perfect square numbers greater than 100 or nearest to 100 = 121 = $(11)^2$

Perfect square numbers greater than 200 or nearest to 200

= 196 = $(14)^2$

The numbers from $(11)^2$ to $(14)^2$ are = 121, 144, 169, 196

Therefore, there will be 4 such perfect square numbers between 100 and 200.

134. Find the least number that should be added to 4042 to make it a perfect square.

(a) 41 (b) 54
(c) 64 (d) 58

RRB Group-D – 22/09/2018 (Shift-III)

Ans. (b) : Square root of 4042,

	63
6	4042
+6	36
123	442
	369
	73

Square of 64 = 64 × 64 = 4096

Hence, the required number = 4096–4042 = 54

By adding 54, the number 4042 will be come perfect square.

135. Divide the number 137592 by the smallest number that leaves no remainder and quotient is a perfect cube. Find the cube root of the quotient.

(a) 8 (b) 2
(c) 4 (d) 6

RRB Group-D – 05/12/2018 (Shift-II)

Ans. (d)

$137592 = 2\times2\times2\times3\times3\times3\times7\times7\times13$

Hence, it is clear that, 137592 leaves no remainder when divided by $7\times7\times13 = 637$

And quotient 216 will be a perfect cube.

$216 = 2\times2\times2\times3\times3\times3$

Hence, the required cube root = $2\times3 = 6$

136. A positive number exceed its positive square root by 30. Find the number.

(a) 16 (b) 36
(c) 25 (d) 49

RRB NTPC 02.04.2016 Shift : 3

Ans : (b) Let the number be x, then-

$x = \sqrt{x} + 30$

$x - 30 = \sqrt{x}$

Squaring in both side-

$(x-30)^2 = (\sqrt{x})^2$

$x^2 + 900 - 60x = x$

$x^2 - 60x - x + 900 = 0$

$x^2 - 61x + 900 = 0$

$x^2 - 36x - 25x + 900 = 0$

$x(x-36) - 25(x-36) = 0$

$(x-36)(x-25) = 0$

$x - 36 = 0$ or $x - 25 = 0$

$x = 36$ or $x = 25$

According to condition, 25 is not more that its square root, which does not follow the condition.

Hence, the required number will be x = 36.

137. What smallest number should be added to the sum of squares of 15 and 14, so that the resulting number is a perfect square?

(a) 17 (b) 20
(c) 11 (d) 9

RRB NTPC 29.03.2016 Shift : 1

Ans : (b) $15^2 + 14^2 = 225 + 196 = 421$

Let the number to added be x,

$421 + x = 441$

$\Rightarrow x = 441 - 421 = 20$

Hence, the required number = 20

138. Calculate the sum of squares of numbers from 1 to 9.

(a) 284 (b) 285
(c) 385 (d) 380

RRB NTPC 27.04.2016 Shift : 1

Ans : (b) The sum of squares of first n numbers

$$= \frac{n(n+1)(2n+1)}{6}$$

$\therefore$ the sum of squares from 1 to 9 will be-

$$= \frac{9(9+1)(18+1)}{6} = \frac{9\times10\times19}{6} = 285$$

139. Calculate the sum of squares of number from 1 to 10.

(a) 384 (b) 285
(c) 385 (d) 380

RRB NTPC 30.04.2016 Shift : 2

Ans : (c) The sum of squares of first n numbers

$$= \frac{n(n+1)(2n+1)}{6}$$

the sum of squares of the numbers from 1 to 10 will be-

$$= \frac{10(10+1)(20+1)}{6} = \frac{10\times11\times21}{6} = 385$$

140. How many of the factors of 256 are perfect squares?

(a) 5 (b) 3
(c) 6 (d) 4

RRB ALP & Tec. (20-08-18 Shift-II)

Ans : (a) Perfect square factors of 256 = 1, 4, 16, 64, 256
Hence, the required number of perfect square factors = 5

141. Find the least number which should be added to 7864, to make it a perfect square.

(a) 61 (b) 57
(c) 71 (d) 79

RRB Group-D – 11/12/2018 (Shift-III)

Ans : (b) When 57 is added to 7864, to make it 7921 which is a perfect square of 89.
Hence, it is clear that adding 57 to 7864 will make the number is perfect square.

142. The number 4050 becomes a perfect square when multiplying by a positive integer. Find the square root of the number.

(a) 95 (b) 80
(c) 90 (d) 85

RRB Group-D – 01/10/2018 (Shift-III)

Ans : (c) $\because 4050 = 2\times\overline{3\times3}\times\overline{3\times3}\times\overline{5\times5}$
Hence, number 4050 becomes a perfect square when multiplied by 2 = $4050\times2 = 8100$
$\therefore$ The required square root of the number 8100
$= 2\times3\times3\times5 = 90$

143. Which of the following numbers is a perfect square?

(a) 0.09 (b) 8.1
(c) 0.025 (d) All

RRB NTPC 29.03.2016 Shift : 2

Ans : (a) $0.09 = (0.3)^2$
Hence, only 0.09 is a perfect square number.

144. Calculate- $\sqrt{30+\sqrt{30+\sqrt{30+\sqrt{30+\sqrt{...}}}}}$**?**

(a) 5 (b) $\sqrt{30}$
(c) 6 (d) 5.8

RRB Group-D – 24/10/2018 (Shift-I)

Ans : (c)

$$\sqrt{30+\sqrt{30+\sqrt{30+\sqrt{30+\sqrt{...}}}}} = x$$

$x = \sqrt{30+x}$
Squaring in both side,
$x^2 = 30 + x$
$x^2 - x - 30 = 0$
$x^2 - 6x + 5x - 30 = 0$
$x(x-6) + 5(x-6) = 0$
$(x+5)(x-6) = 0$
$x + 5 = 0$
$x = -5$ (invalid)
$x - 6 = 0$
$\boxed{x = 6}$ (valid)

Type - 8

145. Which of these numbers has the highest number of divisors?

(a) 156 (b) 240
(c) 172 (d) 200

RRB JE - 23/05/2019 (Shift-I)

Ans : (b)
$156 = 2^2 \times 3^1 \times 13^1 = (2+1)(1+1)(1+1) = 12$ (divisor)
$240 = 2^4 \times 3^1 \times 5^1 = (4+1)(1+1)(1+1) = 20$ (divisor)
$172 = 2^2 \times 43^1 = (2+1)(1+1) = 6$ (divisor)
$200 = 2^3 \times 5^2 = (3+1)(2+1) = 12$ (divisor)
Hence, It is obvious that the number of the divisors of 240 is more.

146. Find the unit digit in given factor of $(3451)^{51} \times (531)^{43}$.

(a) 6 (b) 4
(c) 1 (d) 9

RRB RPF-SI -11/01/2019 (Shift-I)

Ans : (c) The given expression $(3451)^{51} \times (531)^{43}$
According to the question it is clear that the unit digit of 3451 and 531 is 1, so the unit digit of their product will also be 1.

147. How many multiples of $2^8 \times 3^2 \times 5^3 \times 7^5$ are even numbers?

(a) 288 (b) 168
(c) 576 (d) 464

RRB Group-D – 06/12/2018 (Shift-II)

Ans. (c) : The number of factors of $2^8 \times 3^2 \times 5^3 \times 7^5$ = $(8+1)(2+1)(3+1)(5+1) = 648$
$\therefore$ The number of even factors (multiples) = 648 – The number of total odd factors
$= 648 - \{(2+1)(3+1)(5+1)\}$
$= 648 - \{3 \times 4 \times 6\}$
$= 648 - 72 = 576$

148. How many factors of 729 are perfect squares?

(a) 5 (b) 4
(c) 3 (d) 2

RRB Group-D – 01/10/2018 (Shift-I)

Ans. (c) : The factors of 729,

3	729
3	243
3	81
3	27
3	9
3	3
	1

Perfect squares $= \overline{3\times3}\times\overline{3\times3}\times\overline{3\times3}$

Hence, total 3 factors of 729 (9,9,9) are perfect squares.

149. How many multiples of $2^9 \times 3^5 \times 5^4 \times 7^6$ are odd numbers?

(a) 288 (b) 144
(c) 210 (d) 140

RRB Group-D – 06/12/2018 (Shift-III)

Ans. (c) : The required odd multiple number

$= (5+1) \times (4+1) \times (6+1)$
$= 6 \times 5 \times 7 = 210$

150. Find the last digit of 213^6 ?

(a) 6 (b) 3
(c) 7 (d) 9

RRB Group-D – 05/12/2018 (Shift-II)

Ans. (d) The unit digit of 213^6

$213^6 = (213^4 \times 213^2)$
$1 \times 9 = 9$

151. The smallest natural number, by which 216 should be multiplied, so that the number of factors of the product is odd.

(a) 4 (b) 6
(c) 12 (d) 8

RRB Group-D – 11/12/2018 (Shift-I)

Ans. (b) :

The number of multiples of $(216 = 2^3 \times 3^3)$ is:

$= (3+1)(3+1) = 4 \times 4 = 16$ (even)

The smallest natural number, by which 216 should be multiplied, so that the number of factors of the product is odd = 6

$\therefore$ Required number of multiples in $216 \times 6 = 2^4 \times 3^4$

$= (4 + 1)(4 + 1) = 25$

152. What is the unit digit of $[4523^{1632} \times 2224^{1632} \times 3225^{1632}]$

(a) 1 (b) 0
(c) 4 (d) 5

RRB NTPC 18.01.2017 Shift : 3

Ans : (b) $\left[(4523)^{1632} \times (2224)^{1632} \times (3225)^{1632}\right]$

$\Rightarrow (3)^4 \times (4)^4 \times (5)^4$

$81 \times 256 \times 625$

$1 \times 6 \times 5$

$30 \Rightarrow \boxed{0}$

153. Calculate the total prime factors in the product of $\{(8)^{10} \times (9)^7 \times 7^8\}$

(a) 45 (b) 54
(c) 52 (d) 65

RRB NTPC 18.04.2016 Shift : 2

Ans : (c) $(8)^{10} \times (9)^7 \times 7^8$

$= \left((2)^3\right)^{10} \times \left((3)^2\right)^7 \times (7)^8$

$= 2^{30} \times 3^{14} \times 7^8$

Hence, the total prime factors = 30+14+8 = 52

154. Calculate the total prime factors in the product of $\{(16)^7 \times (27)^6 \times 5^9\}$

(a) 28 (b) 43
(c) 55 (d) 56

RRB NTPC 16.04.2016 Shift : 2

Ans : (c) Total prime factors $\{(16)^7 \times (27)^6 \times 5^9\}$

$= \left(2^4\right)^7 \times \left(3^3\right)^6 \times 5^9$

$= 2^{28} \times 3^{18} \times 5^9$

$= 28 + 18 + 9 = 55$

155. Find the unit digit in the product of $(4211)^{102} \times (361)^{52}$

(a) 3 (b) 1
(c) 4 (d) 7

RRB NTPC 16.04.2016 Shift : 3

Ans : (b)

The required unit digit in $(4211)^{102} \times (361)^{52}$

$\Rightarrow (1)^{102} \times (1)^{52} = 1 \times 1 = 1$

156. Find the unit digit in. $(1234)^{102} + (1234)^{103}$

(a) 2 (b) 4
(c) 0 (d) 1

RRB NTPC 28.04.2016 Shift : 2

Ans : (c) Given expression: $(1234)^{102} + (1234)^{103}$

The unit digit,

$= (4)^{102} + (4)^{103}$
$= (4^2)^{51} + (4^2)^{51} \times 4^1$
$= (16)^{51} + (16)^{51} \times 4^1$
$= 6 + 6 \times 4$
$= 6 + 24 = 30$

Hence, the unit digit will be 0.

Type - 9

157. Find the greatest number that will divide 115, 149 and 183 leaving remainders 3, 5, 7 respectively.

(a) 20 (b) 16
(c) 18 (d) 14

RRB JE - 24/05/2019 (Shift-II)

Ans : (b) HCF of the numbers

= (115–3), (149–5) and (183–7)
= 112, 144 and 176
= 16

Hence, the required number is 16.

158. Find the least number which when divided by 5, 6, 7, 8, leaves a remainder 3, and is also a multiple of 9.

(a) 1683 (b) 843
(c) 1677 (d) 1983

RRB JE - 26/05/2019 (Shift-II)

Ans : (a) LCM of 5, 6, 7 and 8 = 840
Required no. = 840k + 3
= 840 × 2 +3 = 1683
[The closest number to 1680 will be 1683 because when dividing by 3 remains 3 as remainder and that is also a multiple of 9.]

159. A number when divided by 10, 9 and 8 separately leaves remainders 9, 8, and 7 respectively. What is the least of such numbers?

(a) 353 (b) 719
(c) 1359 (d) 359

RRB JE - 27/05/2019 (Shift-II)

Ans : (d) (10–9) = (9–8) = (8–7) = 1
∴ The required number = (LCM of 10, 9 and 8) – 1
= 360–1
= 359

160. When a number is divided by a divisor, leaves remainder 24. And twice of the same number is divided by the same divisor, leaves remainder 15. Find the divisor.

(a) 9 (b) 33
(c) 23 (d) 35

RRB RPF-SI -12/01/2019 (Shift-II)

Ans : (b) Let the divided is x, divisor is y and quotient is 1.
Dividend = Divisor × Quotient + Remainder
x = y + 24.....(i)
2x = y + 15.....(ii)
Subtracting equation (i) from equation (ii),
x = –9
Putting the value of x in equation (i),
– 9 = y + 24
$\boxed{y = 33}$

161. Find the least number when divided by 8, 12, 18 leaving remainders 6, 10, 16 respectively.

(a) 74 (b) 144
(c) 146 (d) 70

RRB JE - 30/05/2019 (Shift-I)

Ans : (d) 8 – 6 = 2, 12 – 10 = 2, 18 – 16 = 2
The difference of the digits is equal.
LCM of 8, 12 and 18 = 72
Hence, the required number = 72 – 2 = 70

162. Find the smallest number which, when is reduced 6, it is completely divided by 12, 15, 20 and 27.

(a) 542 (b) 540
(c) 546 (d) 500

RRB Group-D – 24/09/2018 (Shift-II)

Ans : (c) LCM of 12, 15, 20 and 27,
12 = 2 × 2 × 3
15 = 3 × 5
20 = 2 × 2 × 5
27 = 3 × 3 × 3
LCM = 2 × 2 × 3 × 5 × 3 × 3
= 540
Hence, the required number = 540 + 6 = 546

163. Find the greatest number which on dividing 3050 and 5200 leaves remainders 7 and 9 respectively.

(a) 149 (b) 111
(c) 153 (d) 179

RRB NTPC 19.04.2016 Shift : 1

Ans : (d)
3050 – 7 = 3043
5200 – 9 = 5191
3043)5191(1
3043
2148)3043(1
2148
×895)2148(2
1790
×358)895(2
716
179)358(2
358
×××
Hence, the required number = HCF of 3043 and 5191 = 179.

164. When a number is divided by 627 leaves remainder 43. Find the remainder when the same number is divided by 19.

(a) 5 (b) 18
(c) 13 (d) 7

RRB NTPC 12.04.2016 Shift : 1

Ans : (a) 627 is divisible by 19.
Hence, the required remainder
19) 43 (2
38
×5
The remainder will be 5.

165. N is a whole number when, divided by 6 leaves the remainder 4. Find the remainder when 2N is divided by 6.

(a) 4 (b) 8
(c) 2 (d) शून्य

RRB NTPC 28.04.2016 Shift : 1

Ans : (c) Let the quotient be "a" when N is divided by 6.

$\therefore$ N = 6a + 4…….(i)

By equation (i) ×2

2N = 2 × 6a + 8

2N = 12a + 6 + 2

2N = 6 (2a +1) + 2

Hence, the required remainder will be 2.

Type - 10

166. 18 by adding to a two digit number and the digits of the number are interchanged. The product of the digits is '8'. Find the number.

(a) 42 (b) 18

(c) 32 (d) 24

RRB JE - 27/06/2019 (Shift-I)

Ans : (d) Let the unit digit = x.

tens digit = y

The number = 10y + x

xy = 8 ---(i)

According to the question,

10y + x + 18 = 10x + y

9x – 9y = 18

x – y = 2

From equation (i)– $x = \frac{8}{y}$

$\therefore \quad \frac{8}{y} - y = 2$

$8 - y^2 = 2y$

$y^2 + 2y - 8 = 0$

$y^2 + 4y - 2y - 8 = 0$

y (y + 4) –2 (y + 4) = 0

(y –2) (y + 4) = 0

y = 2

Putting the value of y in equation (i)-

x × 2 = 8

x = 4

Number = 10y + x = 10 × 2 + 4 = 24

167. The sum of the digits of a two digit number is 10. When the digits are interchanged is less 36 than the number. Find the changed number.

(a) 82 (b) 73

(c) 37 (d) 28

RRB RPF Constable -17/01/2019 (Shift-III)

Ans : (c) Let the number = 10 x + y

According to the question,

x + y = 10 …….(i)

The number obtained by interchanging digits = (10y + x)

According to the question

(10 x+y) – (10y + x) = 36

$\Rightarrow$ 9x – 9y = 36

x – y = 4 ……(ii)

By adding equation (i) and (ii),

2x = 14

x = 7

$\therefore$ y = 3

Hence, the changed number (10y + x) = 10 × 3 +7 = 37

168. The sum of a two digit number and the number made by interchanging its digits is 132. If the difference of the digits is 4, find the number.

(a) 37 (b) 84

(c) 73 (d) 62

RRB RPF-SI -16/01/2019 (Shift-III)

Ans : (b) Let the tens digit of the number is x and the unit digit is y.

So, the number = 10x + y,

According to the question,

x – y = 4.....(i)

and, $10x + y + 10y + x = 132$

$11x + 11y = 132$

$x + y = 12$.....(ii)

from equation (i) and (ii),

x – y = 4

x + y = 12

2x = 16

x = 8, y = 4

Hence, the required number = $10x + y = 10 \times 8 + 4 = 84$

169. The sum of the digits of a two digit number is 12. The new number formed when the digits are interchanged is 18 more than the original number. What is the original number?

(a) 39 (b) 48

(c) 75 (d) 57

RRB Group-D – 26/11/2018 (Shift-III)

Ans : (d) Let the tens digit of the number is x and the unit digit is y.

x + y = 12 .....(i)

Hence, the two digit number = 10x + y

the number obtained by the digit to the interchange according to the question,

10 y + x = 10x + y + 18

9y – 9x = 18

9x – 9y = –18

$\boxed{x - y = -2}$.....(ii)

By adding equation (i) and (ii),

x + y = 12

x – y = –2

2x = 10

$\boxed{x = 5}$

$\boxed{y = 7}$

Hence, the required number = 10x+y = 10×5+7 = 57

170. The sum of the digits of a two digits number is 9. Also nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

(a) 19 (b) 18

(c) 28 (d) 30

RRB Group-D – 05/11/2018 (Shift-III)

Ans. (b) : Let the tens digit is x and the unit digit is y.
$\therefore$ The number = 10x + y
According to the first condition,
$x + y = 9$ .....(i)
According to the second condition,
$(10x + y) \times 9 = (10y + x) \times 2$
$90x + 9y = 20y + 2x$
$88x = 11y$
$y = 8x$
Putting the value of y in equation (i),
$x + 8x = 9$
$x = 1$
Putting the value of x in equation (i),
$1 + y = 9$
$y = 8$
Hence, the number = 10x + y
$= 10 \times 1 + 8 = 18$

171. The sum of the digits of a two digit number is 11. The new number formed when the digits interchanged is 45 less than the original number. Find the original number.
(a) 92 (b) 56
(c) 65 (d) 83
RRB Group-D – 15/10/2018 (Shift-III)

Ans. (d) : Let the tens digit of the number is a and the unit digit is b.
So, the number = 10a + b
According to the question,
$a + b = 11$ .....(i)
According to the question,
$10b + a = 10a + b - 45$
$9a - 9b = 45$
$a - b = 5$ .....(ii)
Adding equation (i) and (ii),
$a + b = 11$
$a - b = 5$
$2a = 16$
$a = 8$
Putting the value of a in equation (i),
$8 + b = 11$
$b = 3$
Hence, the required number = $10 \times 8 + 3 = 83$

172. The difference between a number of two digits and the new number formed when the digits are interchanged is 45. Find the difference between the two digits.
(a) 4 (b) 5
(c) 6 (d) 7
RRB NTPC 05.04.2016 Shift : 2

Ans : (b) Let the tens digit be x.
And the unit digit be y.
The number = 10x + y
According to the question,
$(10x + y) - (10y + x) = 45$
$9x - 9y = 45$
Hence, the required difference will be $x - y = 5$

173. The sum of the digits of a two digit number is 11. If the digits are interchanged, the number decreases to 63. Find the number.
(a) 83 (b) 92
(c) 29 (d) 38
RRB NTPC 04.04.2016 Shift : 3

Ans : (b) Let the tens digit be x and the unit digit be y of the number.
$\therefore$ The number = $10x + y$
According to the question-
$x + y = 11$ ..... (i)
and $10y + x = 10x + y - 63$
$9x - 9y = 63$
$x - y = 7$ ..... (ii)
By adding equation (i) and (ii)
$2x = 18 \Rightarrow x = 9,\ y = 2$
Hence, the number = $10x + y = 10 \times 9 + 2 = 92$

174. The sum of the digits of a two digit number is 9. When 27 is added to the number, the digits are interchanged. Find the number.
(a) 45 (b) 36
(c) 18 (d) 27
RRB NTPC 03.04.2016 Shift : 1

Ans : (b) Let the unit digit be x in the two digit number.
Then,
According to the question,
the tens digit = $9 - x$ And the number = $10(9 - x) + x$
Again According to the question,
$10(9 - x) + x + 27 = 10x + 9 - x$
$\Rightarrow 90 - 10x + x + 27 = 9x + 9$
$\Rightarrow 90 + 27 - 9 = 18x$
$\Rightarrow 18x = 108$
$x = 6$
Then, the number $= 10(9-x) + x$
$= 10(9-6) + 6 = 36$

175. The sum of the digits of a two digit number is 13. If those digits are interchanged, the number gets decreased by 27. Find the changed number.
(a) 85 (b) 76
(c) 67 (d) 58
RRB NTPC 02.04.2016 Shift : 1

Ans : (d) Let the tens digit is x,
the unit digit = $13 - x$
$\therefore$ The number = $10 \times x + (13 - x)$
According to the question,
$10 \times (13 - x) + x = 10 \times x + (13 - x) - 27$
$130 - 10x + x = 10x + 13 - x - 27$
$18x = 144$
$x = 8$
$\therefore$ The changed number,
$= 10 \times (13 - x) + x$
$= 10 \times (13 - 8) + 8$
$= 10 \times 5 + 8 = 58$

176. The sum of a two digit number is 9. The number is reduces from 45, when the digits are interchanged, find the changed number.
(a) 45 (b) 72
(c) 63 (d) 27
RRB NTPC 02.04.2016 Shift : 2

Ans : (d) Let the tens digit be = x
And the unit digit be = y
Number = 10 x + y
Then, x + y = 9 ……..(1)
According to the question, interchanging the place-
(10 x + y) – (10 y + x) = 45
9x – 9y = 45
x – y = 5 ..... (2)
From, equation (1) + (2)
2x = 14 ⇒ x = 7
From, equation (1),
y = 9–7 = 2
Hence, The required number = 10y+x = 10×2+7 = 27

177. The sum of digits of a two-digit number is 10. When the digits are reversed, the number decreases by 54. Find the new number.
(a) 73 (b) 28
(c) 82 (d) 37
RRB NTPC 02.04.2016 Shift : 3

Ans : (b) Let the tens digit of the number is x and the unit digit is y.
∴ The number = 10x + y
According to the question,
$x + y = 10$ ---------(i)
and $10x + y = 10y + x + 54$
$\Rightarrow 9x - 9y = 54 \Rightarrow x - y = 6$ -----(ii)
By adding equation (i) and (ii),
$2x = 16 \Rightarrow x = 8, y = 2$
Hence, the new number
$= 10y + x = 10 \times 2 + 8 = 28$

178. The sum of digits of a two-digit number is 10. When the digits are inter changed, the number increases by 18. Find the number.
(a) 46 (b) 64
(c) 19 (d) 28
RRB NTPC 29.03.2016 Shift : 1

Ans : (a) Let the tens digit of the number is x and the unit digit is y.
∴ The number $= 10x + y$
According to the question-
$x + y = 10$ ----------(i)
$10x + y = 10y + x - 18$
$9x - 9y = -18$
$x - y = -2$ ----------(ii)
By adding equation (i) and (ii) -
$2x = 8 \Rightarrow x = 4$, $y = 6$
Hence, The required number $= 10 \times 4 + 6 = 46$

179. A two digits number and its digits the sum of the number formed by interchanging its digits, is 99. Find the number if the difference of the digits is 3.
(a) 27 (b) 63
(c) 45 (d) 54
RRB NTPC 10.04.2016 Shift : 3

Ans : (b) Let the unit digit be y and the tens digit be x.
∴ The number = 10x + y
According to the question,
(10x + y) + (10y + x) = 99
11x + 11y = 99
x + y = 9 ……(i)
x – y = 3 ……..(ii)
By adding equation (i) and (ii),
2x = 12
x = 6
From equation (i), y = 3
∴ The required number = 10x+y = 10×6+3 = 60+3= 63

180. The difference of a number and the number formed by interchanging its digits, is 54. Find the difference between the digits of the number.
(a) 6 (b) 7
(c) 8 (d) 9
RRB NTPC 06.04.2016 Shift : 1

Ans : (a) Let the tens digit of the number be x and the unit digit be y.
∴ The number = 10x + y
According to the question-
$(10x + y) - (10y + x) = 54$
$10x + y - 10y - x = 54$
$9x - 9y = 54$
$x - y = \frac{54}{9} = 6$
Hence, the difference between the digits is 6.

181. The sum of the digits of a two digit number is 5. When the digits are reversed the number decreases by 9. Find the changed number.
(a) 32 (b) 23
(c) 41 (d) 14
RRB NTPC 28.04.2016 Shift : 3

Ans : (b) Let the tens digit of the number be x and the unit digit be y.
∴ the number = 10x + y
According to first condition,
x + y = 5 ……(i)
The obtained number after reversing the digits = 10y+x
According to the question,
(10x+y) – (10y+x) = 9
⇒ 9x – 9y = 9
⇒ x – y = 1 .....(ii)
By adding equation (i) and (ii),
2x = 6
x = 3
From equation (ii)
3 – y = 1
y = 3–1 = 2
Hence, the changed number = 10y + x
= 10 × 2 + 3 = 23

Type - 11

182. What is the difference between the place and face values of '5' in the number 3675149?
(a) 5000 (b) 4995
(c) 495 (d) 4990
RRB JE - 23/05/2019 (Shift-I)

Ans : (b) The place value of 5 in the number 3675149 = 5×1000 = 5000
And the face value of 5 = 5
The difference = 5000–5 = 4995

183. What is the place value of 8 in 634785?
(a) 8 (b) 80
(c) 800 (d) 80,000
RRB RPF Constable -20/01/2019 (Shift-I)

Ans : (b) The place value of 8 in 634785 = 8 ×10 = 80

184. Find the sum of the face value and place value of 6 in the number 206743?
(a) 6749 (b) 12743
(c) 6006 (d) 12
RRB Group-D – 28/11/2018 (Shift-I)

Ans : (c) The face value of 6 in the number 206743 = 6
And the place value of 6 = 6 × 1000 = 6000
Face value + Place value = 6 + 6000 = 6006

185. The difference between two place values of 3 in 935071360 is-
(a) 29999700 (b) 29999701
(c) 2999600 (d) 29999400
RRB Group-D – 23/10/2018 (Shift-I)

Ans. (a) : The place value of 3 in 935071360,
935071360
↓ ↓→ = 3×100=300
↓→ =3×10000000=30000000
The required difference = 30000000–300 = $\boxed{29999700}$

186. Calculate the sum of the face value and the place value of 7 in the number 3728456.
(a) 700007 (b) 0
(c) 7 (d) 700000
RRB Group-D – 01/10/2018 (Shift-I)

Ans. (a) : The place value of 7 = 7 × 100000 = 700000
The face value of 7 = 7.
Hence, the required sum = 700000 + 7 = 700007

187. Find the face value of 4 in 145.390.
(a) 40,000 (b) 4
(c) 140,000 (d) 45
RRB NTPC 04.04.2016 Shift : 2

Ans : (b) The face value of any digit is the same digit.
in the number 145.390
the face value = 4

188. Find the difference between the place value and face value of the digit 9 in the number 229301?
(a) 9292 (b) 8991 (c) 0 (d) 220
RRB NTPC 03.04.2016 Shift : 2

Ans : (b) The place value of 9 in the number 229301 = 9 × 1000 = 9000
And the face value of 9 = 9
Hence, the required difference = 9000- 9 = 8991

189. What is the difference between the place value and face value of 3 in 273965?
(a) 2035 (b) 3962
(c) 2997 (d) 0
RRB ALP & Tec. (31-08-18 Shift-II)

Ans. (c) : In the number 273965,
The place value of 3 = 3 x 1000 = 3000
And the face value = 3
Hence, the required difference = 3000 – 3 = 2997

190. The difference between the place values of '4' and '2' in the number 833749502 is:
(a) 49998 (b) 30098
(c) 39098 (d) 39998
RRB ALP & Tec. (10-08-18 Shift-II)

Ans : (d) The given number = 833749502
The place value of 2 = 2
The place value of 4 = 4 x 10000 = 40000
Hence, the required difference = 40000 – 2 = 39998

Type - 12

191. Solve: $1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+.....$
(a) 2 (b) 1/50
(c) 3 (d) 1/22
RRB JE - 23/05/2019 (Shift-I)

Ans : (a)
$1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+.....$
This is a geometrical progression-
$a=1,\ \ r=\frac{1}{2}$
Let the sum be S.
$S=\frac{a}{1-r}$, $S_\infty=\frac{1}{1-\frac{1}{2}}$
$S_\infty=\frac{1}{\frac{1}{2}}=2$ $\boxed{S_\infty=2}$

192. In a school picnic group, 2/9th part were adults and the number of children was more than adults by 95. How many children were present there?
(a) 95 (b) 133
(c) 190 (d) 103
RRB JE - 27/06/2019 (Shift-I)

Ans : (b) Let the total number = x
The number of adults = $x\times\frac{2}{9}=\frac{2x}{9}$
The number of children = $x-\frac{2x}{9}=\frac{9x-2x}{9}=\frac{7x}{9}$
$\frac{7x}{9}-\frac{2x}{9}=95$
$\frac{7x-2x}{9}=95$

$\frac{5x}{9} = 95$

$x = 171$

Hence, the number of children $= \frac{7x}{9} = \frac{7}{9} \times 171 = 133$

193. Find the value of 52-|8-20|=

(a) 45 (b) 40
(c) 65 (d) 64

RRB RPF Constable -18/01/2019 (Shift-I)

Ans : (b) The given value = $52 - |8 - 20|$

$= 52 - |-12|$

$|-A| = A$ The value of Mod is always +ve.

Hence, the required value $= 52 - 12 = 40$

194. If one dozen of apples weigh 1.8 kg, then find the number of apples of three boxes whose total weight is 23.25 kg.

(a) 280 (b) 155
(c) 465 (d) 215

RRB RPF-SI -13/01/2019 (Shift-I)

Ans : (b) Total weight = 23.25 kg

One apples's weight $= \frac{1.8}{12}$ kg

The required number of apples,

$= \frac{\text{Total weight}}{\text{1 apple's weight}} = \frac{23.25 \times 12}{1.8} = 155$

Number of apples is box is = 155

195. Pick out the set that forms the factors of 36.

(a) (2, 3, 4, 6, 9) (b) (2, 3, 4, 6)
(c) (2, 3, 4, 6, 9, 12, 18) (d) (2, 3, 4, 6, 9, 12)

RRB JE - 27/05/2019 (Shift-III)

Ans : (c) All the factors of 36 = 1, 2, 3, 4, 6, 9, 12, 18, 36
Hence, the required set that is formed by the factors of 36 will be = (2, 3, 4, 6, 9, 12, 18)

196. The difference of two numbers is 5. If their product is 336, find the sum of the numbers.

(a) 21 (b) 37
(c) 28 (d) 51

RRB JE - 26/06/2019 (Shift-I)

Ans. (b) Let the numbers be x and y respectively.

$x - y = 5$ (i)

$xy = 336$ (ii)

$(x + y)^2 = (x-y)^2 + 4xy$

From equation (i) and (ii),

$(x + y)^2 = (5)^2 + 4 \times 336$

$(x + y)^2 = 25 + 1344$

$(x + y)^2 = 1369$

$(x + y) = \sqrt{1369}$

$x + y = 37$

Hence, the required sum of the numbers =37

197. The square of a number is 3 more than twice the number. What is the possible number.

(a) 1 or 3 (b) 1 or -3
(c) -1 or -3 (d) -1 or 3

RRB Group-D – 15/10/2018 (Shift-I)

Ans : (d) Let the number be x

According to the question,

$x^2 = 2x + 3$

$x^2 - 2x - 3 = 0$

$x^2 - 3x + x - 3 = 0$

$x(x - 3) + 1(x - 3) = 0$

$(x - 3)(x + 1) = 0$

$x - 3 = 0$

$x = 3$

$x + 1 = 0$

$x = -1$

Hence, the possible number is -1 or 3.

198. $\left(\frac{3}{10} + \frac{8}{15}\right)$ is directly proportional to-

(a) $\frac{11}{10}$ (b) $\frac{11}{15}$
(c) $\frac{6}{5}$ (d) $\frac{3}{15}$

RRB Group-D – 02/11/2018 (Shift-I)

Ans. (c) $\left(\frac{3}{10} + \frac{8}{15}\right)$ is directly proportional to-

$\frac{9+16}{30} = \frac{25}{30} = \frac{5}{6}$

$\frac{5}{6}$ is directly proportional to $= \frac{1}{\frac{5}{6}} = \frac{6}{5}$

199. Subtract 64.37 out of 1000.03 and add the resultant obtained from it to the sum of 3.4 and 7.56. What will be its value?

(a) 948.62 (b) 944.62
(c) 945.62 (d) 946.62

RRB Group-D – 08/10/2018 (Shift-III)

Ans : (d) According to the question,

$1000.03 - 64.37 = 935.66$

$935.66 + (3.4 + 7.56) = 935.66 + 10.96 = 946.62$

200. Seema got ₹ 50 from her father and purchased toffee from ₹ 15. Her mother gave her ₹ 30 while her brother took ₹ 42 from her. How much money did she have left?

(a) ₹ 23 (b) ₹ 24
(c) ₹ 20 (d) ₹ 25

RRB Group-D – 23/09/2018 (Shift-II)

Ans : (a) Total sum of money that Seema have left
= 50-15+30-42 = 80-57 = ₹ 23

201. Geeta weighs 11.235 kg. Her sister weighs 1.4 times her weight. Find the total weight of both.

(a) 15.729 kg (b) 25.964 kg
(c) 26.964 kg (d) 28.964 kg

RRB NTPC 29.03.2016 Shift : 1

Ans :(c) Geeta's weight $= 11.235$ kg

$\therefore$ The weight of Geeta's sister

$= 11.235 \times 1.4 = 15.729$ kg

The total weight of both is

$= 11.235 + 15.729 = 26.964$ kg